

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2019

EE 1213

(Electrical Circuits and Electronics)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define unilateral and bilateral circuits. Identify the following elements as Active or Passive, and explain the reasons: 06

(i) Resistor

(ii) Inductor

(iii) Capacitor

(iv) Diode

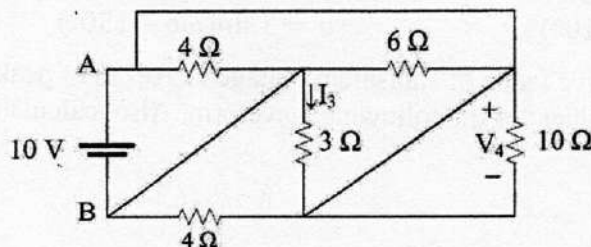
(v) Transistor

(vi) Memristor

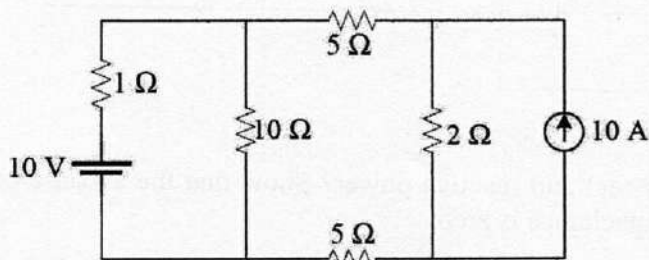
- 1(b). Differentiate between AC and DC. Show that the time period T of a DC signal is infinite. 05

- 1(c). What is maximum power transfer theorem? Explain when a load will receive maximum power? 10

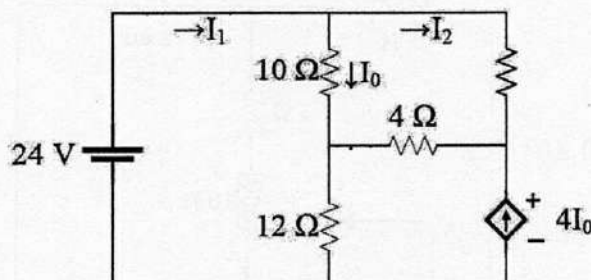
- 1(d). State Kirchhoff's voltage and current law. Explain them briefly. Find the current I_3 and voltage V_4 for the following network. 14



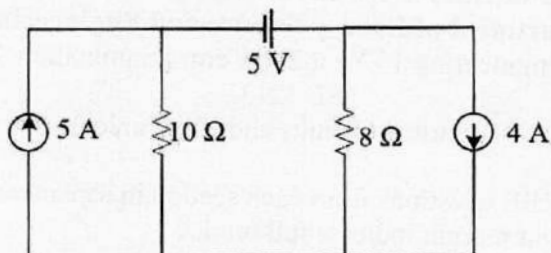
- 2(a). State and explain super position theorem. What are the conditions to apply to this theorem? 13
Find the voltage across 10 Ω resistor using superposition theorem as shown in the following figure.



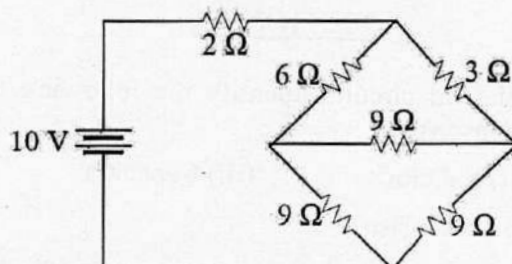
- 2(b). Find the current I_0 for the following circuit using mesh analysis. 12



- 2(c). Explain source conversion technique. Find the current through $8\ \Omega$ for the following network using source conversion technique. 10



- 3(a). Find the total resistance and current for the following network: 10



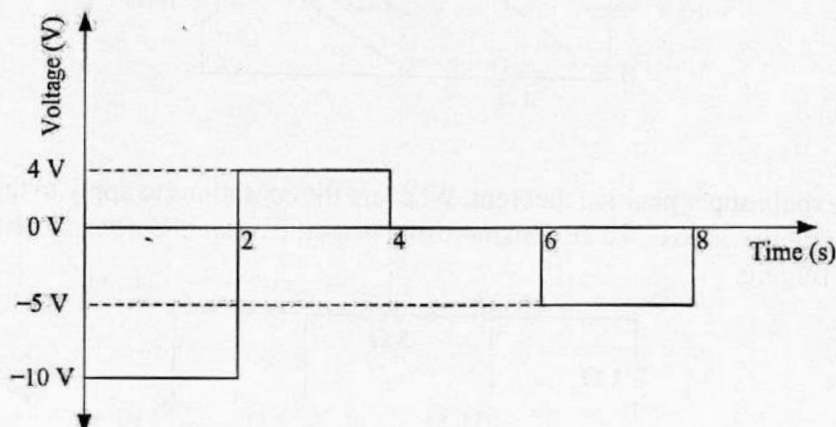
- 3(b). Define (i) instantaneous value, (ii) peak amplitude, (iii) periodic waveform, (iv) average value, (v) RMS value, and (vi) form factor. 06

- 3(c). What is the phase relationship between the sinusoidal waveform of each of the following sets. Also, draw their phasor diagram and instantaneous waveform. 10

(i) $i = 2 \cos(\omega t + 10^\circ)$
 $v = 3 \sin(\omega t - 10^\circ)$

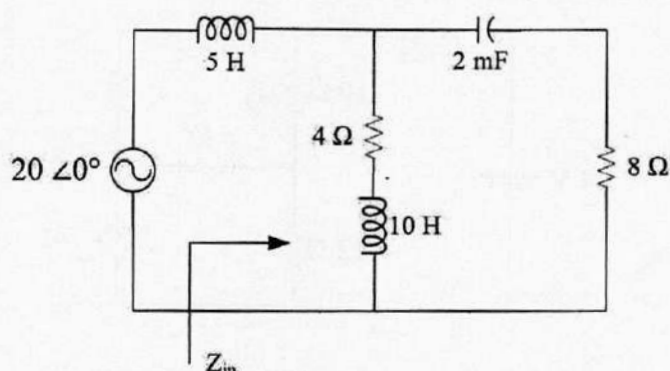
(ii) $i = -2 \cos(\omega t - 60^\circ)$
 $v = 3 \sin(\omega t - 150^\circ)$

- 3(d). Prove that, the effective value of sinusoidal voltage is 70.7% of peak voltage. Find the average and RMS values for the following waveform. Also, calculate the form factor. 09

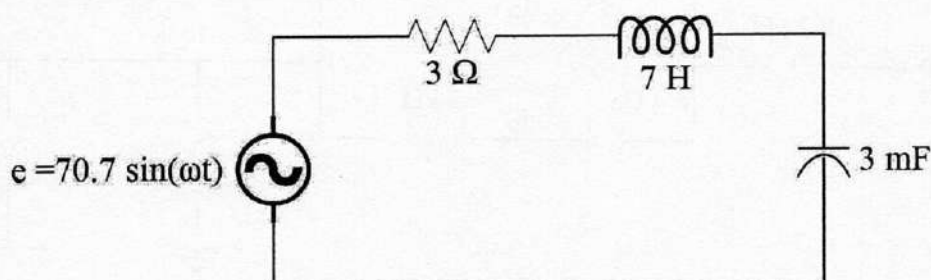


- 4(a). What is meant by real and reactive power? Show that the average power consumed by inductance and capacitance is zero. 10

- 4(b). Find the input impedance of the following network. Note that the circuit operates at $\omega = 50\text{ rad/s}$. 10



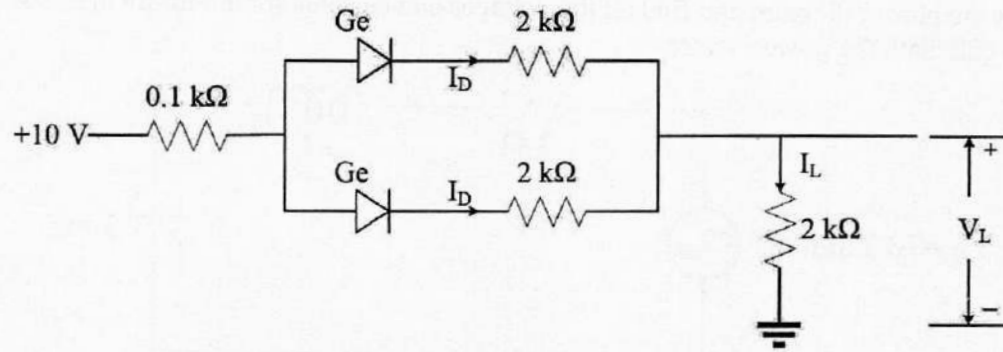
- 4(c). What is meant by resonance in RLC circuit? Drive the expression of resonant frequency and Q -factor of parallel resonance circuit. 07
- 4(d). Draw the phasor diagram and find all the voltages and currents for the following network. Also calculate the powerfactor. 08



SECTION – B

- 5(a). Define semiconductor and classify it. Why do we need extrinsic semiconductor? 03
- 5(b). Derive mass action law with neat sketch and explain its significance. 12
- 5(c). Draw the energy band diagram of intrinsic and extrinsic semiconductors showing fermi level. Show that for n -type semiconductor, $E_C - E_F < E_F - E_V$, where the symbols indicate the usual meaning. 10
- 5(d). Find the resistance of a $1 \mu m^3$ pure silicon crystal. Given data atomic concentration in silicon is $5 \times 10^{22} Cm^{-3}$, intrinsic carrier concentration of silicon is $1 \times 10^{10} Cm^{-3}$, electron and hole mobility in silicon crystal are $1350 Cm^2 V^{-1}s^{-1}$ and $450 Cm^2 V^{-1}s^{-1}$, respectively. 10
- 6(a). For an ideal pn junction, derive the equation of built-in potential across the depletion region with neat sketches. 13
- 6(b). Discuss the dependence of electron concentration of a n -type semiconductor on temperature. 09
- 6(c). Draw and explain the energy band diagram for a pn junction at (i) open circuit, (ii) forward bias, (iii) reverse bias, and (iv) thermal generation of electron hole pairs. 10
- 6(d). Calculate the built-in potential for an abrupt $Si p^+n$ junction diode at a temperature $27^\circ C$ having donor doping concentration in n side and acceptor doping concentration in p side are $10^{16} Cm^{-3}$ and $5 \times 10^{18} Cm^{-3}$, respectively. 03
- 7(a). Write short notes on: 10
(i) Avalanche break down (ii) Zener break down
- 7(b). For a particular semiconductor material $N_C = 1.5 \times 10^{18} cm^{-3}$, $N_V = 1.3 \times 10^{19} cm^{-3}$ and $E_g = 1.43 eV$ at $T = 300 K$. 10
(i) Determine the intrinsic carrier conductance of the semiconductor at $T = 300 K$.
(ii) Find the position of the intrinsic Fermi level with respect to the center of the band gap.

- 7(c). For the circuit shown in following figure, find the current I_D through each diode, load current I_L and voltage across the load V_L . 08



- 7(d) Discuss about binary ripple counter with diagram. 07
- 8(a). Mention the differences between combinational and sequential circuit. 05
- 8(b). Explain the operation of *RS* flipflop using *NAND* latch. 07
- 8(c). What is universal gate? Make the basic gates *AND*, *OR*, *NOT*, and *NOR* using universal gate. 08
- 8(d). Draw a 4-bit bidirectional shift register with parallel load, and mention its function table. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2019

Hum 1213

(Technical English)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Make sentence with each of the following structures using the words given in brackets. 14
- (i) Subject + Verb + Adverb of place. (play as verb)
 - (ii) There + Verb + Subject + Adverb of place. (work as verb)
 - (iii) What + Subject + Verb + Adverb of manner + Verb + Adjective complement. (say and is as verb)
 - (iv) Subject + Verb + That + Subject + Verb + Adverb of place. (know and succeed as verb)
 - (v) Subject + Verb + Object + What + Subject + Verb. (entitle and reflect as verb)
 - (vi) Subject + Verb + Not only + Object + But also + Object. (eat as verb)
 - (vii) Subject + Verb + Neither + Adjective complement + Nor + adjective complement. (is as verb)
- 1(b). Make use of the following words in sentence as asked in brackets. 12
- Pin (as verb), grass (as verb), cloud (as verb), chair (as verb), love (as adj.), leather (as adj.)
- 1(c). Write two antonyms for each of the following words and use the antonyms in sentence. 09
- (i) Mobilize, (ii) Sincere, and (iii) Complex.
- 2(a). Make WH question with each of the underlined word/words in the following sentences. 14
- (i) Liza can play cricket.
 - (ii) Labu meets us everyday.
 - (iii) He wants to see the sight.
 - (iv) He comes here to experience the thing.
 - (v) Mim tackles the situation with diligence.
 - (vi) The pond is loft deep.
 - (vii) Simu has been studying in this versity for four years.
- 2(b). Change the following words as directed and make sentences with the changed form. 12
- (i) conclude (adj.), (ii) vivacity (adv.), (iii) actively (verb), (iv) habituate (noun), (v) methodize (adv.), and (vi) historically (verb).
- 2(c). Make new words with the following prefixes and suffixes and use the new words in sentences. 09
- Anbi —, Bi —, Neo —, — ness, — ian, — let

- 3(a). Transform the following sentences as directed. 14
- (i) Nobody insulted him. (affirmative)
 - (ii) The sunbeam is wonderful to look at. (exclamatory)
 - (iii) Since he worked hard, he topped the list. (simple)
 - (iv) I saw him singing. (complex)
 - (v) April is the hottest month in Bangladesh. (positive)
 - (vi) When I woke up, it was dark. (compund)
 - (vii) we should read the daily newspaper. (imperative)
- 3(b). Complete the following sentences with clauses. 12
- (i) is true. (noun clause)
 - (ii) There is no meaning in (noun clause)
 - (iii) I know the boy (adj. clause)
 - (iv) The girl came here. (adj. clause)
 - (v), I would sing to cheer you. (adv. clause or condition)
 - (vi) I shall support you (adv. clause or conserion)
- 3(c). Make sentences with the following phrases and idioms: 09
- Queer fish, Ring donn the curtain, Square meal, Snake in the grass, Talking shop, Laeonic speech.
- 4(a). Correct the following sentences: 14
- (i) The hen has given two eggs.
 - (ii) She gave up her studies two years before.
 - (iii) Each of them were happy.
 - (iv) Are you alone or married?
 - (v) Please excuse me to be so late.
 - (vi) I have an account on this bank.
 - (vii) Zillur's is close to nine.
- 4(b). Make use of the follöwing modals in sentence as asked in braekets. 12
- (i) Shall. (To ask for opinion about something from someone else)
 - (ii) Should. (To express natural in ference)
 - (iii) Should + be + bare form of verb + ing. (To express duty at present)
 - (iv) May + be + bare form of verb + ing. (To express guess about present situation)
 - (v) Dare. (To express indulgence)
 - (vi) Need. (To express opinion about something)
- 4(c). Define with example: (i) present participle, (ii) infinitive, and (iii) gerund. 09

SECTION – B

- 5(a). Read the following passage carefully and answer the questions that follows. 20
- People use metal detectors to find metals. Metal detectors make magnetic waves. These waves go through the ground. The waves change when they hit metal. Then the device beeps. This lets the person with the device know that metal is close.
- The first metal detectors were meant to help miners. They were big, costly and used a lot of power. However, they did not work well. People kept trying to make them better. Thus they got smaller. Now they are light, cheap and works better. That is why people bring them to the beach. They can look for rings in the water and phones in the sand. Metal detectors help them find these.
- Metal detectors also protect people. They help to keep guess out of some places. They are in airports, court houses and schools. They help guards look for weapons. Guards use special wands to find metal on a person. These devices save lives in other ways too. During wars, people plant bombs in the ground. When the wars end, these places become unsafe for people. Metal detectors are used to find bombs and remove them.
- These devices also make clothes safer. While sewing, needles break from time to time and get stuck in the clothes. They would poke people trying them on. They don't though. That's because our clothes are scanned for metal. Hence, metal detectors make the world a safer place.
- (i) How do metal detectors save lives?
 - (ii) Why would buying an early metal detector have been disappointing?
 - (iii) Explain how metal detectors make buying clothes as safer experience
 - (iv) In what other ways can you use a metal detector to save life?
- 5(b). Make a précis of the above passage mentioned in question no. 5(a) with a suitable title. 15
- 6(a). Amplify the following idea: 20
- “Sweet are the uses of adversity”
- 6(b). Write a paragraph on ‘perseverance’. 15
- 7(a). Suppose the Department of ESE has arranged a special course for all the students to enhance their knowledge and experience to take proper career decision. Prepare a memo on the basis of the given information. 15
- 7(b). George Washington University is seeking for a Security Trainee who would like to participate in energy efficiency programs. Prepare your CV and apply for this post. 20
- 8 Write a composition on any one of the followings: 35
- (i) Advantages and disadvantages of alternative energy
 - (ii) Animated Movies

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2019

ESE 1205

(Thermodynamics for Energy Engineering)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What does it mean by macroscopic and microscopic view points of thermodynamics? Explain with example. 07
- 1(b). Define pure substance and phase. Discuss the condition based on which air can be treated as a pure substance. 08
- 1(c). Discuss about thermodynamic equilibrium. 10
- 1(d). A closed, rigid tank filled with water, initially at 2 MPa, a quality of 80%, and a volume of 0.5 m³ is cooled until the pressure is 0.4 MPa. Show the process of water on a sketch of the T-v diagram and evaluate the heat transfer. 10
- 2(a). State Zeroth law of thermodynamics. Which thermodynamic property is based on this law? 05
- 2(b). A mass of 8 kg gas expands within a flexible container so that the P-v relationship is of the form $Pv^{1.2} = \text{constant}$. The initial pressure is 1000 kPa and the initial volume is 1 m³. The final pressure is 5 kPa. If specific internal energy of the gas decreased by 40 kJ/kg, find the heat transfer in magnitude and direction. 12
- 2(c). Draw an energy mind map. Define the different terms in the map. 12
- 2(d). Write the definition of the following terms with figure if necessary: 06
(i) Open system
(ii) Thermodynamic process
(iii) Total energy of a system
- 3(a). State the first law of thermodynamics. Write down the energy rate equation for an open system and mention the meaning of each term. 10
- 3(b). Derive the first and second Tds equations. Also, show that for an incompressible fluid, change in specific entropy is given by the integral $\int_{T_1}^{T_2} \frac{c(T)}{T} dT$, where $c(T)$ is the specific heat. 10
- 3(c). At steady state, water enters the waste heat recovery steam generator at 290 kPa, 104 °C, and exits at 276 kPa, 160 °C, shown in a figure in the next page. The steam is then fed into a turbine from which it exits at 7 kPa and a quality of 90%. Air from an oven exhaust enters the steam generator at 182 °C, 100 kPa, with a volume flow rate of 2 m³/s, and exits at 138 °C, 100 kPa. Ignore all stray heat transfer with the surroundings and all kinetic and potential energy effects. If the power developed is valued at 7 taka per kWh, do you recommend implementation of this waste-heat recovery system? Provide supportive calculations. 15

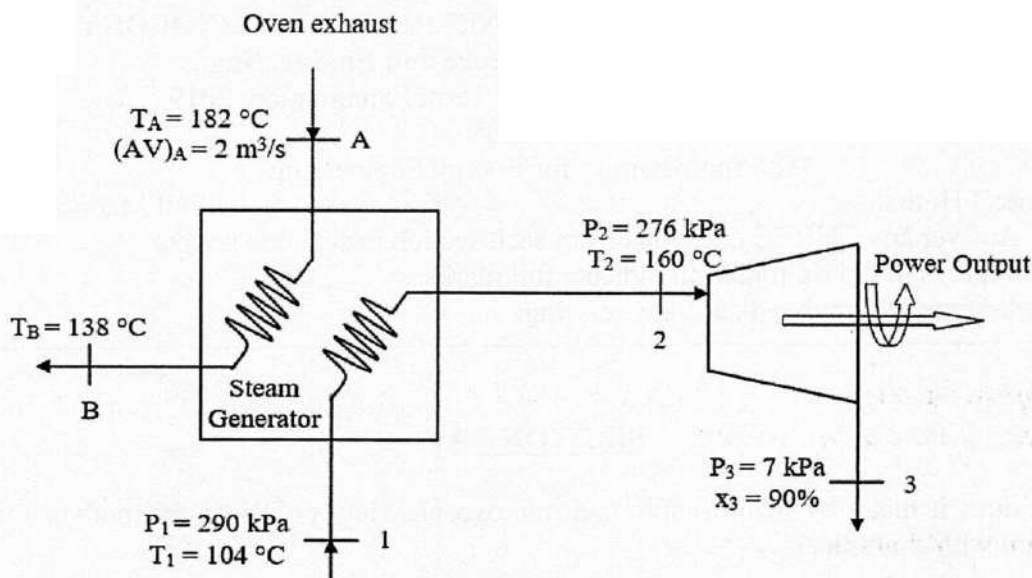


Figure for 3(c)

- 4(a). State 'Kelvin-Plank' statement of second law of thermodynamics and discuss its mathematical form. 07
- 4(b). State 'State postulate' of a simple compressible system. What is its significance? 05
- 4(c). Define exergy. Prove that maximum theoretical value for the work of a system is given by: 15

$$W_c = (U - U_0) + P_0(V - V_0) - T_0(S - S_0) + KE + PE$$
- 4(d). Write short notes on the following terms: 08
 (i) Carnot efficiency (ii) Irreversible processes
 (iii) Clausius inequality (iv) Thermal reservoir

SECTION – B

- 5(a). What is mean temperature of heat addition? Show that the cycle efficiency is a function of the mean temperature of heat addition. 08
- 5(b). From T-s diagram, show that the deviation of an actual Rankine cycle from the ideal one. Explain in brief. 06
- 5(c). Why ideal Rankine cycle is modified to reheat Rankine cycle and regenerative Rankine cycle? 05
- 5(d). In a Rankine cycle, the steam enters the turbine at 4 MPa and 400 °C. Heat is rejected in the condenser at 27 °C. Calculate the thermal efficiency of the cycle. Also calculate the thermal efficiency if the isentropic efficiency of the turbine is 85% and the isentropic efficiency of the pump is 87%. 16
- 6(a). What is air standard cycle? Write down the air standard assumptions of an actual cycle. 08
- 6(b). From the derivation, show that the efficiency of Diesel cycle is less than that of the Otto cycle for the same compression ratio. 07
- 6(c). An engine operating on an air standard Dual cycle, the pressure and temperature at the start of compression are 0.1 MPa and 27 °C. At the end of compression, pressure reaches a value of 2.3 MPa. 430 kJ of heat is supplied per kg of air during constant heating and pressure becomes 0.3 MPa at the end of isentropic expansion. Estimate the pressure and temperature at all other cardinal points. Also, calculate the cycle efficiency. 20

- 7(a). With a neat schematic diagram, describe the working principle of vapor absorption refrigeration system. 10
- 7(b). Define the followings: 04
 (i) Gas mixture (ii) Gravimetric analysis (iii) Volumetric analysis
 (iv) Ideal gas
- 7(c). State Dalton's law of partial pressure. Show that for a gas mixture, 06
Partial pressure = Mole fraction × Total pressure
- 7(d). A mixture of ideal gases consists of 3.5 kg of N_2 , 5.5 kg of CO_2 at pressure 300 kPa and temperature 200 °C. Find: (i) mass fraction and mole fraction, (ii) equivalent molecular weight of the mixture, (iii) volume and density of the mixture, (iv) C_p and C_v of the mixture. Assume γ for CO_2 and N_2 are 1.29 and 1.40 respectively. 15
- 8(a). Why dry bulb temperature is always greater than wet bulb temperature? 05
- 8(b). Why a wet cloth dries slower in summer than in winter? 05
- 8(c). Prove that the relative humidity ϕ is given by: 09

$$\phi = \frac{\mu}{1 - (1 - \mu)(P_s/P)}$$
 where, μ = degree of saturation, P_s = saturation pressure of vapor in moist air, P = barometric pressure of moist air.
- 8(d). Atmospheric air at 1.0132 bar has a dry bulb temperature of 35 °C and wet bulb temperature of 31 °C. Compute the: (i) partial pressure of water vapor, (ii) specific humidity, (iii) relative humidity, (iv) dew point temperature, (v) enthalpy, and (vi) specific volume. 16

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2019

Math 1213

(Differential Equation and Co-ordinate Geometry)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Write the relations to transform from spherical polar to Cartesian co-ordinates. Also obtain the spherical polar co-ordinates of Cartesian co-ordinate $(-1, -1, -\sqrt{2})$. 10
- 1(b). What conic does the equation $8x^2 + 4xy + 5y^2 - 16x - 14y + 13 = 0$ represent? Reduce it to standard form. 13
- 1(c). Determine the angle through which the axes must be rotated to remove xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$. Also, find the transformation equation. 12
- 2(a). Define direction cosine and direction ratios of a line. A line makes angles α, β, γ , and δ with four diagonal of a cube. Show that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$. 15
- 2(b). Using vector form, find the perpendicular distance of a corner of a unit cube from a diagonal not passing through it. 10
- 2(c). Find the vector equation of a straight line passing through the origin and parallel to a given vector $\vec{b}$. 10
- 3(a). Find the vector equation of a plane passing through two points a and b , and is parallel to the given vector $\vec{c}$. 10
- 3(b). Find the equation of the plane through $(5, 2, -3)$ and perpendicular to each of the plane $\vec{r} \cdot (2\vec{i} - \vec{j} + 2\vec{k}) = 0$ and $\vec{r} \cdot (\vec{i} + 3\vec{j} - 5\vec{k}) + 3 = 0$. 10
- 3(c). A plane meets the coordinate axes at A, B, C such that the centroid of the triangle ABC is at the point (α, β, γ) . Find the equation of the plane. 15
- 4(a). If possible, determine the shortest distance between the lines: 15
- $$\frac{x+3}{2} = \frac{y+5}{3} = \frac{z+7}{-3} \text{ and } \frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$$
- Comment whether the given straight lines are coplanar or not.
- 4(b). Find the vector equation of a straight line which passes through the point $(1, 2, -3)$ and parallel to the straight line joining the points $(2, 3, -1)$ and $(3, 5, 2)$. 12
- 4(c). Find the equation of the line passing through the origin and parallel to the vector $3\vec{i} - 2\vec{j} + 4\vec{k}$. 08

SECTION – B

- 5(a). Define order and solution of an ordinary differential equation. Find the general solution of $(1 - x^2) \frac{dy}{dx} = 1 + xy$ 12
- 5(b). Solve : $y dx - x dy = \sqrt{x^2 + y^2} dx$ 11
- 5(c). Determine the general solution of $(x^3 - 2y^2) dx + 2xy dy = 0$ 12

- 6(a). A body originally at 80 °C cools down to 60 °C in 20 minutes, the temperature of the air being 40 °C. What will be the temperature of the body after 40 minutes from the original? 12
- 6(b). Solve: $\frac{d^2y}{dx^2} + 4y = 3e^x \cos 2x + 4 \sin 2x$ 12
- 6(c). Solve: $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2x + x^2$ 11
- 7(a). By the method of variation of parameters solve: 12
- $$\frac{d^2y}{dx^2} - y = \frac{2}{1 + e^x}$$
- 7(b). Define Laplace transform. Find the Laplace transform of $e^{3t}(2 \cos 5t - 3 \sin 5t)$. 12
- 7(c). State the convolution theorem. Find the Laplace transform of the function 11
- $$f(t) = \begin{cases} E \sin \omega t & 0 < t < \pi/\omega \\ 0 & \pi/\omega < t < 2\pi/\omega \end{cases}$$
- 8(a). Find the inverse Laplace transform of $\left\{ \frac{3s+7}{(s+1)(s-3)} \right\}$ using Heaviside's expansion formula. 11
- 8(b). Find the inverse Laplace transform of $\left\{ \frac{3}{s^2(s+2)} \right\}$ using convolution theorem. 11
- 8(c). Solve: $\frac{d^2y}{dt^2} + 9y = \cos 2t$ 13
- when, $y(0) = 1$ and $y(\pi/2) = -1$ using Laplace transform.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2019

Ch 1213

(Chemistry II)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

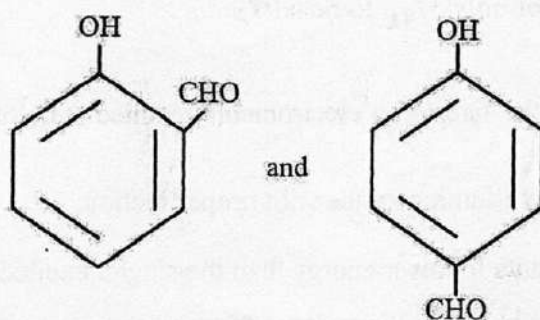
SECTION – A

- | | | |
|-------|---|----|
| 1(a). | How does increase of temperature lead to an increase in reaction rate? | 10 |
| 1(b). | The rate of a chemical reaction mixture can be determined by measuring electrical conductivity of that mixture – explain. | 11 |
| 1(c). | Write down some advantages of Transition State theory over Collision theory of chemical reaction. | 09 |
| 1(d). | In certain first order reaction, half life of the reactant is decomposed in 10,000 seconds. How long will it take for only $\frac{1}{10}$ to be left? | 05 |
| | | |
| 2(a). | How can you express the fate of an electronically excited state of a molecule? Explain with a suitable diagram. | 10 |
| 2(b). | Discuss an artificial chemiluminescence with proper reaction. | 09 |
| 2(c). | Why is triplet excited state in lower energy than the singlet excited state? | 06 |
| 2(d). | What is meant by quantum yield? Write down some reasons responsible for high quantum yield. | 10 |
| | | |
| 3(a). | Why ethyne is acidic in nature whereas ethene and ethane are neutral? Explain with the help of hybridization concept. | 10 |
| 3(b). | What is hybridization? What are the distinguishing features of sp^3 and sp^2 orbitals? | 08 |
| 3(c). | Discuss the lock and key mechanism of enzyme catalyzed reaction. | 10 |
| 3(d). | Draw the potential energy diagram of enzyme catalyzed and uncatalyzed reaction. | 07 |
| | | |
| 4(a). | Explain enantiomers and diastereomers with suitable examples. | 08 |
| 4(b). | Derive the rate equation for second order reaction. | 10 |
| 4(c). | Write down two basic differences between spectroscopy and photochemistry. | 08 |
| 4(d). | Define the following terms: | 09 |
| | (i) Photon (ii) Electrophile (iii) Nucleophile | |

SECTION – B

- | | | |
|-------|--|----|
| 5(a). | What is meant by sacrificial anode? Explain the anodic protection of corrosion. | 07 |
| 5(b). | What is meant by passivity of metal? Explain the active-passive behavior of metal with diagram. | 12 |
| 5(c). | What is meant by salting out effect? Explain the effect of dissolved salt on the rate of underwater corrosion. | 10 |
| 5(d). | Write a short note on pitting corrosion. | 06 |

- 6(a). Deduce Langmuir adsorption isotherm and discuss this equation for the limiting condition of very low and very high pressure. 12
- 6(b). Write down the differences between physical and chemical adsorption. 06
- 6(c). Derive Michaelis-Menten equation for enzyme catalyzed reaction. 10
- 6(d). Discuss the applications of adsorption with suitable example. 07
- 7(a). What is meant by electromagnetic radiation? How do the wavelength and frequencies of different types of electromagnetic radiation vary? 07
- 7(b). State Beer-Lambert's law. How can this law be used for the estimation of an analyte? 09
- 7(c). Sketch and explain about different electronic transition on irradiation of UV-visible radiation. 10
- 7(d). Explain the following terms: 09
 (i) Blue shift (ii) Auxochromes (iii) Quantization of energy
- 8(a). Explain the origin of line spectra and band spectra. 10
- 8(b). How can you distinguish between the following compounds using IR spectra? 10



- 8(c). What is finger print region in IR spectra and why is it so called? 07
- 8(d). Explain about pretreatment of surface. 08

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